SCHOOL OF MECHANICAL ENGINEERING - MECHATRONICS DEPARTMENT

CDRA

Animatronics · Octopus-Inspired · Cable Mechanism · 6-DOF · Assistive Technology

Development of a Cable-Driven Robotic Arm *for Wheelchair*

PRESENTED BY

Rameshwar Cambow
Sourabh Devrukhkar

CONTENTS

01 **Introduction & Motivation**

Animatronics · octopus-inspired design · assistive purpose

02 **Joint Design**

Bionic vertebrate structure · 6-DOF · cable actuation

03 **Gripper Design**

Cable-actuated 4-subsystem claw · 3 fingers

04 **Design Method**

Continuum robot · bio-inspired mechanism

05 **Control System**

Hand gesture + joystick · Arduino + DC motors

06 **Conclusion & Future Scope**

Autonomy · AI · camera integration

INDEX

INTRODUCTION

01

Background

OCTOPUS-INSPIRED

🐙 Muscular Hydrostat

Arms inspired by octopi's unique muscle structure capable of twisting, extending and changing stiffness freely in any direction. Provides the bio-mechanical basis for infinite-DOF motion.

ANIMATRONICS

🤖 Multidisciplinary Field

Combining anatomy, puppetry and mechatronics pioneered by Walt Disney in the 1960s. PC and human control both workable. Movement actuators duplicate strong development of figures.

ASSISTIVE PURPOSE

♿ Independence for Users

Enables wheelchair users to board buses and cars independently no external assistance required. Designed for a large population of people of all ages who rely on wheelchairs daily.

6	±45°	7	3
DEGREES OF FREEDOM	BENDING PER 2 JOINTS	DOF WITH END EFFECTOR	GRIPPER FINGERS

The Scale of the Problem

1.3B

People worldwide living with disability

WHO, 2023

75M+

Wheelchair users globally estimated

WHO / Lancet

90%

of wheelchair users in developing countries

ISPO Report

80%

of daily tasks require upper limb assistance

Rehab. Eng. Review

The Access Barrier

Only ~2% of buses and ~5% of urban rail stations in low-income countries are fully wheelchair accessible (WHO 2023). Manual boarding remains the dominant challenge.

Economic Cost

The global assistive technology market was valued at $18.3B in 2022 and is projected to reach $38.7B by 2030 at 9.8% CAGR driven by aging populations (Grand View Research).

Technology Gap

Fewer than 10% of wheelchair users who need assistive robotic arms currently have access to one mainly due to cost ($20K–$80K for commercial systems) and complexity (Frontiers in Robotics, 2022).

Limitations of Conventional Robotic Arms

Mechanical Rigidity

Conventional manipulators are constrained by fixed link structures — limited to pre-defined motion paths with no ability to conform to irregular surfaces or obstacles.

Low DOF

Standard rigid arms typically offer 3–4 DOF. Achieving human-like flexibility requires prohibitively complex and expensive multi-joint structures.

Size and Weight

Fixed-link robotic arms are heavy and bulky — impractical as wheelchair attachments. Commercial systems (e.g., JACO, Manus) weigh 6–8 kg and cost $20K–$80K.

Actuator Overhead

Each DOF in a rigid arm requires a dedicated motor at that joint — adding mass, power consumption, and control complexity exponentially with each added degree.

CONVENTIONAL VS. CDRA

Property	Rigid Arm	CDRA
DOF	3–4	**6–7**
Weight	6–8 kg	**~1–2 kg**
Bending	Fixed joints	**±45°/2j**
Conforms to env	No	**Yes**
Cost	$20K–80K	**Low-cost**
Actuators/DOF	1 per joint	**Tendons**

INTRODUCTION

04

Comparative Technology Review

RIGID LINK ARM

Kinova JACO / Manus

DOF: 6–7 fixed

Weight: 6–8 kg

Cost: $20K–$80K

ADVANTAGES

+ Well-understood mechanics
+ High payload capacity
+ Precise positioning

LIMITATIONS

– Heavy & bulky (6–8 kg)
– Fixed DOF, low flexibility
– $20K–$80K cost

PNEUMATIC SOFT ARM

Festo Bionic Arm

DOF: Multi

Weight: ~0.5 kg

Cost: Medium

ADVANTAGES

+ Very compliant and soft
+ Safe around humans
+ Lightweight

LIMITATIONS

– Needs air compressor
– Slow response time
– Difficult to control precisely

CABLE-DRIVEN ARM

CDRA (This Work)

DOF: 6–7 DOF

Weight: ~1–2 kg

Cost: Low-cost

ADVANTAGES

+ High DOF, infinite conform
+ Remote actuation (light)
+ Bio-inspired flexibility

LIMITATIONS

– Cable wear over time
– Complex routing
– Calibration needed

EXOSKELETON ARM

Ekso / ReWalk Arm

DOF: 3–5

Weight: 5–12 kg

Cost: $100K+

ADVANTAGES

+ Augments human motion
+ Strong force amplification
+ Clinical approval

LIMITATIONS

– Requires active user input
– Very expensive ($100K+)
– Bulky external frame

JOINT DESIGN

The Bionic Vertebrate Structure

DESIGN FEATURES

🦴 Bionic Vertebrate Structure

Inspired by human spinal column — interlocking joints form a strong yet flexible assembly, identical to vertebrae stacking.

🔩 Three-Component Framework

Adaptive arm + claw/gripper + control framework. Longitudinally determined links connect all joints inside the arm.

💡 Lighting Equipment

Manipulator joint equipped with integrated lighting to enable operation in dark or low-visibility conditions.

⚡ Servo-Driven Cables

Driven cables simulate transverse muscle effects same as octopi's arms. Servo engines interface determined wires for precise movement.

6 DEGREES OF FREEDOM

±45° BENDING ANGLE

∞ JOINT TYPE

Inspired by octopus muscular hydrostat

GRIPPER DESIGN

06

Cable-Actuated Gripper System

The gripper has four subsystems and is inspired by the human hand supporting two grasping modes (EG & PG) with high reliability.

1

3-finger claw

Frame

Flexible arm supports a 3-finger claw with holder for mounting on the robotic arm. Built for structural rigidity and easy finger swapping.

2

Embedded cables

Cable Mechanism

Cables run through each finger and connect to arm cables for smooth, coordinated actuation across all joints.

3

0°–180° opening

Control System

Small gears driven by servomotors enable full open/close action. Claw opens up to 180° with complete closure capability.

4

Active bending

Fingers

Each finger embeds a cable through its structure and tip. Bending is actively controlled — works on any terrain or surface.

✋ *Cable-actuated soft fingers preferred over fluid-actuated simpler, swappable, controllable bending radii*

DESIGN METHOD

Mechanism & Adaptive Interface

BIO-INSPIRED CONTINUUM ROBOT

Living organisms evolved over millions of years. Continuum robots like elephant trunks, earthworms and snakes navigate tight regions and work in complex environments. The octopus can simultaneously curve, twist and contract different segments to achieve peristaltic movement.

ADAPTIVE USER INTERFACE

Two techniques: User Modeling (HCI-based) and Adaptive User Interface, which responds dynamically to data stored from previous interactions. Individual profiles describe hand force, word finishing and assistive-type propensities. Accommodates a wide range of disability types.

WHEELCHAIR INTEGRATION

Definite Dynamics Manus Assistive Robotic Manipulator deployed as the arm type. Complete system = 6+3 DOF wheelchair with arm. Augmented with real-time hand gesture and joystick control modes for maximum user accessibility.

ARM TYPE

Manus Assistive

Robotic Manipulator

TOTAL DOF

6 + 3

arm + gripper

CONTROL

Dual

gesture & joystick

ACTUATION

DC Servo

motors + Arduino

DESIGN METHOD

Robotic Arm Integration Overview

DESIGN /TECHNICAL SPECIFICATIONS

09

Full System Specifications

PARAMETER	VALUE	UNIT / NOTE
Degrees of Freedom	6	Arm DOF
DOF with Gripper	7	Arm + end effector
Max Bending Angle	±45°	Per 2 joints
Claw Opening Range	0° – 180°	Full close to full open
Number of Fingers	3	Cable-actuated soft
Servo Motors (Arm)	Multiple	Per joint actuation
Servo Motors (Claw)	4	3 right + 1 left pulley
Frame Material	3D Printed	Lightweight rigid PLA/ABS
Control Modes	2	Joystick + Hand Gesture
Microcontroller	Arduino	Servo & cable interface
Max Payload	~1–2 kg	Estimated (light objects)

MARKET CONTEXT

$38.7B

Assistive Tech market by 2030

Grand View Research 2022

9.8%

Market CAGR 2022–2030

GVR Market Report

< 10%

Access rate for robotic arms

Frontiers in Robotics 2022

CONTROL SYSTEM

Dual Control Modes

Hand Gesture Control

- Optical contraptions detect region & heading of hand movements
- Widely used across modern industries from robotics to consumer electronics
- High-performance real-time system required for accurate response
- Next-generation replacement for joystick improves ergonomics
- Reduces physical burden; hands-free operation possible

Joystick Control

- Atari CX40 joystick left, right, backward, forward commands
- Commands received at transmitter end by receiver module
- Cables from arm connect to DC-powered motor driver circuit
- Motor driver connected to Arduino for signal processing
- Preferred for ease-of-use; more intuitive for most wheelchair users

SIGNAL CHAIN · Joystick / Gesture → Transmitter → Receiver → Motor Driver → Arduino → Servo → Cables → Arm

CONCLUSION

Real-World Impact

Public Transport Access

Only 2% *of buses are wheelchair accessible globally (WHO 2023)*

The CDRA arm enables wheelchair users to grip handrails and assist boarding of buses and trains independently. A direct response to the global public transport accessibility deficit.

Medical & Rehabilitation

70M+ *people need a wheelchair globally, most lack assistive arm access*

Can support post-stroke or paralysis patients in performing reaching and grasping tasks. Cable-driven compliance ensures safe interaction around patients zero rigid-arm impact risk.

Industrial & Hazardous Zones

$6.4B *global service robotics market in healthcare/industrial use (IFR 2023)*

Fixed robots with cable-driven arms are already used in medical and hazardous environments. CDRA expands this to mobile wheelchair-mounted platforms dramatically extending reach.

CONCLUSION & FUTURE SCOPE

12

Outcomes & Next Steps

CONCLUSIONS

- ✓ Cable-driven arm inspired by octopi's arms built on animatronics principles
- ✓ Arms move in infinite directions, mirroring human spinal cord structure
- ✓ Easily controlled via joystick minimal training for end users
- ✓ Designed for the large population of wheelchair users of all ages
- ✓ Enables independent boarding of buses and cars no assistance required

FUTURE SCOPE

F1

HAND GESTURE CONTROL

Replace the joystick

Gesture control will supersede joystick, improving wheelchair ergonomics and enabling more intuitive, hands-free operation.

F2

CAMERA & SCREEN

Enhanced visual feedback

Rear camera for operator navigation; arm-mounted camera for precision gripping and surface detection in all conditions.

F3

AUTONOMOUS OPERATION

AI + sensor fusion

Sensors and image processing enable fully autonomous arm movement and gripping no operator input required.

Thank You

Cable-Driven Robotic Arm for Wheelchair

Animatronics · Octopus-Inspired · Cable Mechanism · 6-DOF · Assistive Technology